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Pett

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(54) **ASPIRATION APPARATUS FOR
INTRAOSSEOUS ACCESS SYSTEM**

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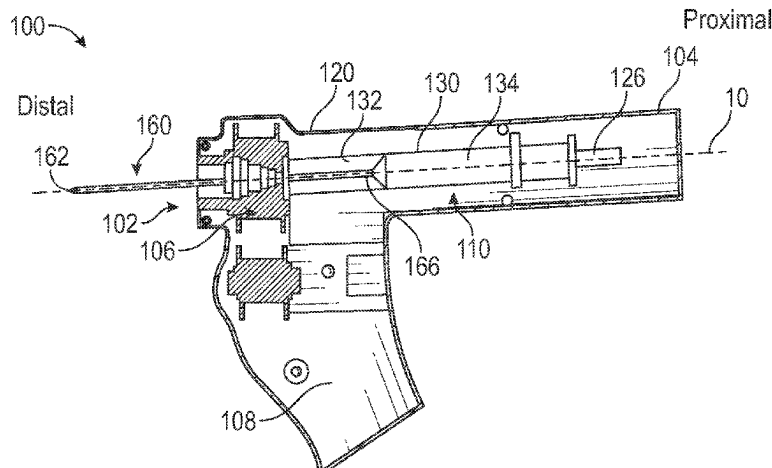
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See application file for complete search history.

(57) **ABSTRACT**

Embodiments disclosed herein are directed to an intraosseous access system configured to confirm access to a medullary cavity. The system includes a driver housing, an access system including a drive train, a needle assembly rotatably coupled to the access system, and an aspiration system. The needle assembly can include a needle defining a lumen and an obturator disposed therein. The aspiration system can include a syringe or a vacutainer, coupled to the obturator. Sliding one of a plunger or a vacutainer proximally can withdraw the obturator from the needle lumen and place a vacuum in fluid communication with the needle lumen to draw a fluid flow therethrough. A portion of the aspiration system can rotate along with the needle assembly. Withdrawing the obturator can place the obturator within the syringe or the vacutainer mitigating needle stick injuries. If

(Continued)



the medullary cavity has not been accessed, the obturator can be replaced.

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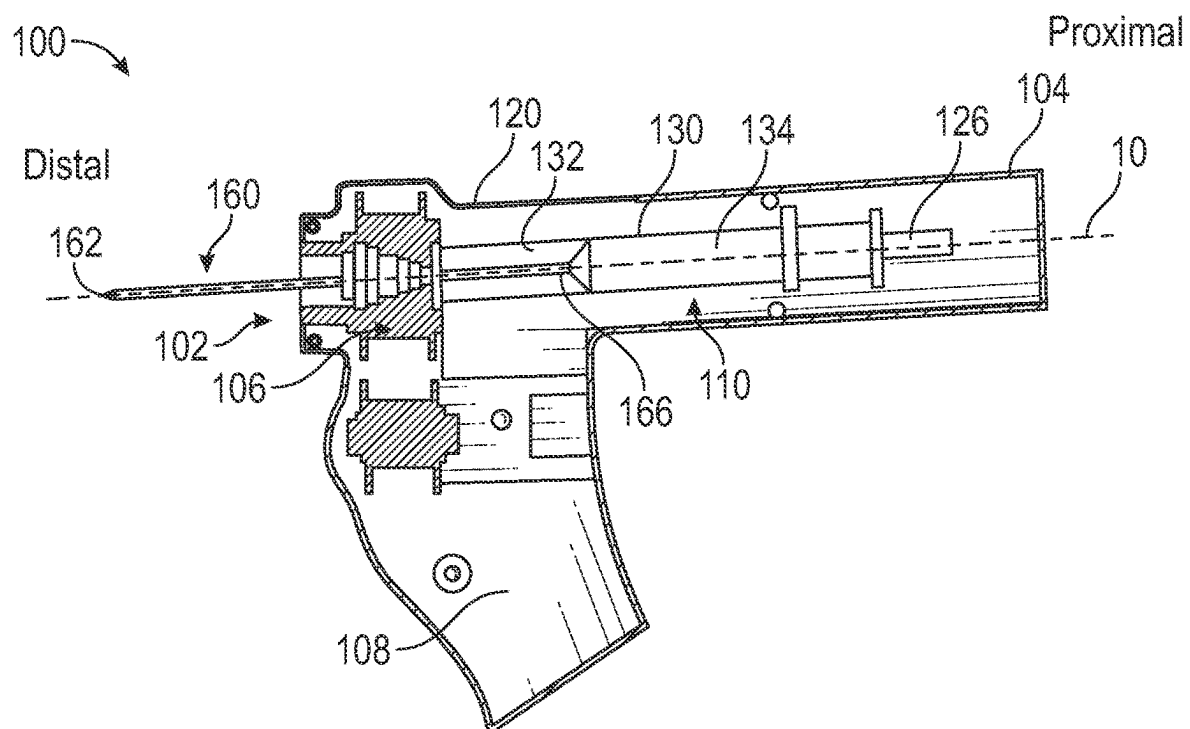


FIG. 1A

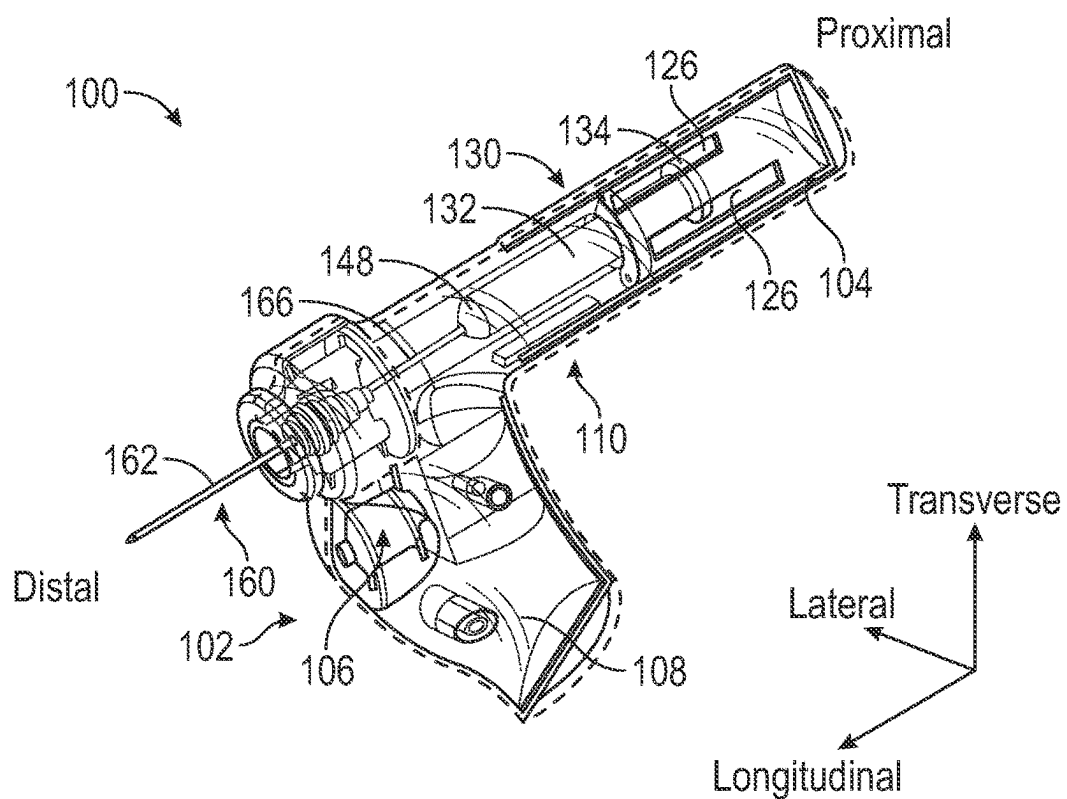


FIG. 1B

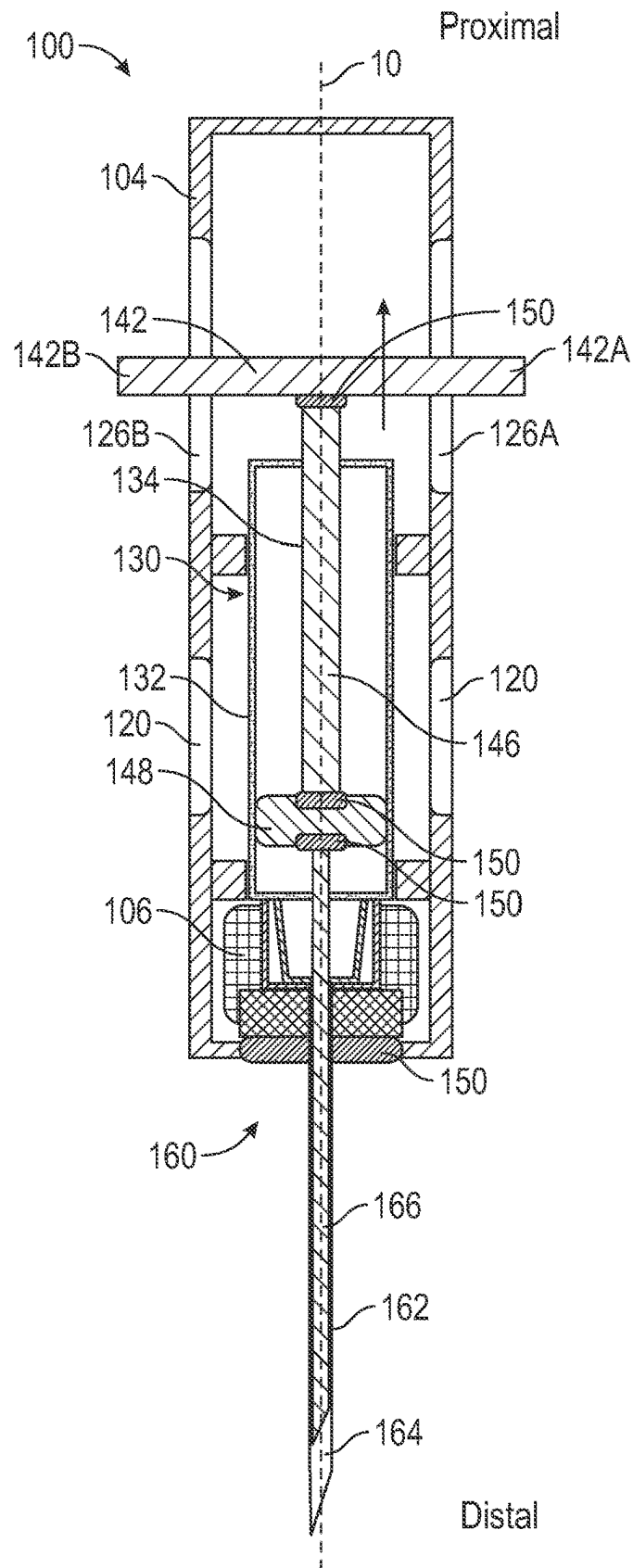


FIG. 2

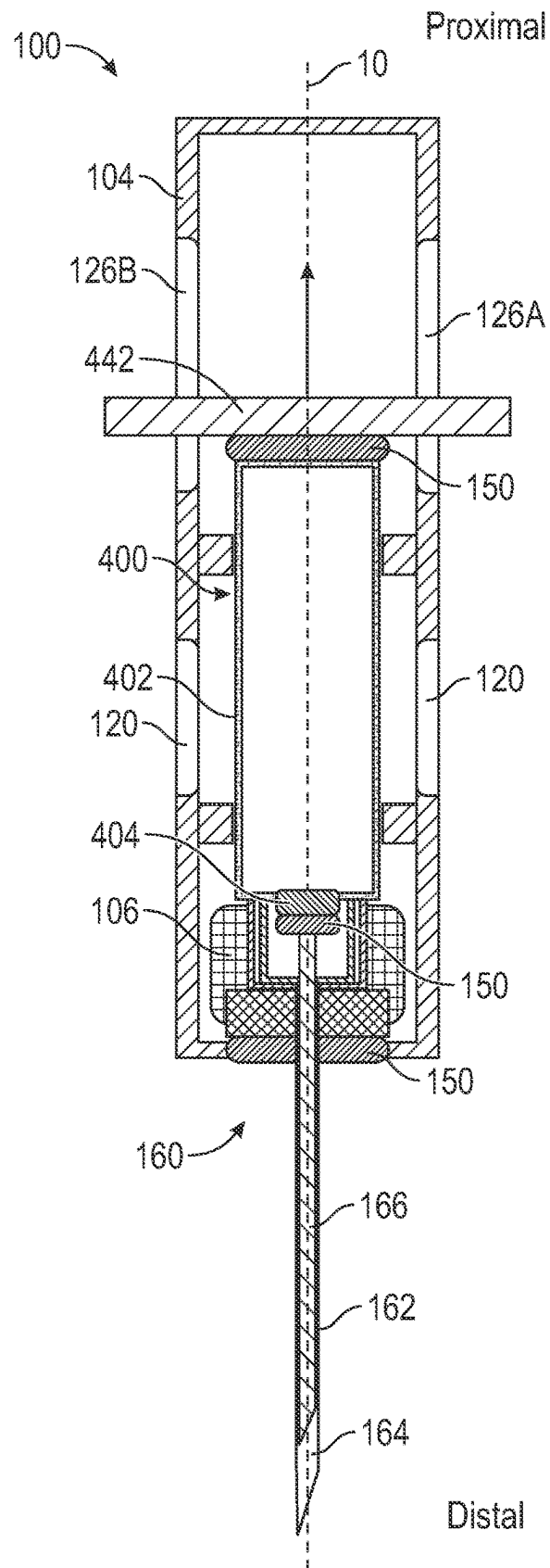


FIG. 3

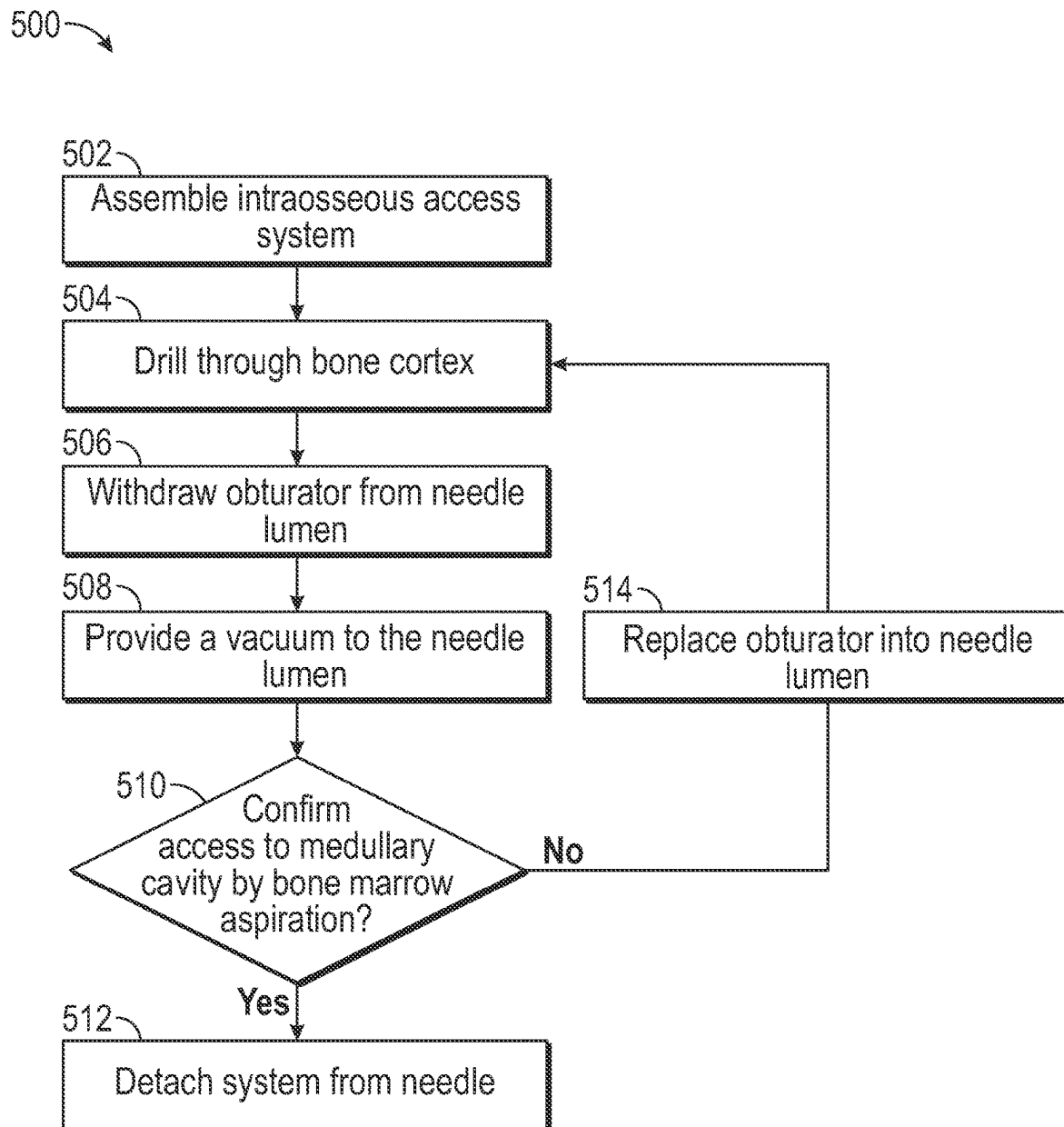


FIG. 4

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ASPIRATION APPARATUS FOR INTRAOSSEOUS ACCESS SYSTEM

PRIORITY

This application claims the benefit of priority to U.S. Patent Application No. 63/076,189, filed Sep. 9, 2020, which is incorporated by reference in its entirety into this application.

BACKGROUND

Intraosseous (IO) access systems are configured to access a medullary cavity of a bone by drilling an access needle through the dense bone cortex and into the medullary cavity. Some IO access systems include a powered drill, or similar device, configured to rotate a needle assembly at high rotational speeds to drill an access needle through the bone cortex. Detecting when the needle has accessed the medullary cavity can be challenging. For example, to confirm access to the medullary cavity, a user must disengage the drill from the needle assembly, withdraw the obturator from a lumen of the needle, attach a syringe to the access needle and aspirate the needle lumen to observe the presence of bone marrow. Failure to aspirate bone marrow may mean that the access needle has not fully accessed the medullary cavity and further drilling is required. However, with the needle assembly disengaged from the drill, and the obturator withdrawn from the needle lumen, the needle cannot be reused and the clinician must withdraw the needle, couple a new needle assembly to the drill, and continue drilling. Further, patency of the original access site may be lost during needle exchange.

Disclosed herein are systems, apparatus and methods directed to address the foregoing problems.

SUMMARY

Disclosed herein is an intraosseous access system including, a housing, a needle assembly extending from a distal end of the housing and including an obturator extending along a longitudinal axis and disposed within a lumen of a needle, an aspiration system disposed within the housing and including one of a syringe or a vacutainer, the aspiration system configured to slide the obturator relative to the needle and selectively provide a vacuum in fluid communication with the needle lumen, and a drive train disposed within the housing and configured to rotate the needle assembly and one of a syringe barrel or a vacutainer canister about the longitudinal axis.

In some embodiments, the drive train includes one of a biasing member, an electric motor, or a battery. In some embodiments, the syringe includes a plunger slidably engaged with the syringe barrel and the housing along a longitudinal axis, the syringe barrel rotatable about the longitudinal axis relative to the plunger. In some embodiments, the intraosseous access system further includes a piston rotatably coupled to one of the plunger, the syringe barrel, or the obturator. In some embodiments, the intraosseous access system further includes a handle coupled to the plunger, a portion of the handle extending through a wall of the housing.

In some embodiments, the vacutainer includes a valve configured to selectively place a vacuum, disposed within the vacutainer canister, in fluid communication with the needle lumen. In some embodiments, the vacutainer further includes an actuator configured to slide the canister along a

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longitudinal axis or actuate the valve between an open and closed position. In some embodiments, the housing further includes an observation window configured to allow a user to observe one of the syringe barrel or vacutainer canister disposed therebelow. In some embodiments, one of the syringe barrel or vacutainer canister can include a transparent material. In some embodiments, the biasing member is one of a flat spring, a coiled spring, or a torsion spring. In some embodiments, a distal tip of the obturator is configured to be withdrawn into one of the syringe barrel or the housing to mitigate accidental needle stick injuries.

Also disclosed is a method for accessing a medullary cavity including, actuating a drive train disposed within a housing, rotating a needle assembly coupled to the drive train and including an obturator disposed within the a lumen of a needle, the needle extending along a longitudinal axis, rotating one of a syringe barrel or a vacutainer canister relative to the housing about the longitudinal axis, withdrawing an obturator from the needle lumen to provide a fluid communication between the needle lumen and one of the syringe barrel or the vacutainer canister, and replacing the obturator within the needle lumen.

In some embodiments, the drive train includes one of a biasing member, a flat spring, a coiled spring, a torsion spring, an electric motor, or a battery. In some embodiments, the method further includes sliding a handle coupled to one of a plunger or the vacutainer canister along a longitudinal axis relative to the housing, the handle remaining in a rotatably fixed position about the longitudinal axis, relative to the housing. In some embodiments, the handle extends through an aperture disposed in a wall of the housing. In some embodiments, the method further includes rotating a piston coupled to a distal end of the plunger, relative to one of the handle or the syringe barrel, the piston providing a fluid tight seal between a wall of the syringe barrel and the piston.

In some embodiments, the method further includes confirming access to the medullary cavity by observing a fluid flow into one of the syringe barrel or the vacutainer through one or more viewing windows disposed in the housing. In some embodiments, the method further includes detaching the needle from the needle assembly. In some embodiments, the method further includes disposing the obturator within one of the syringe barrel or the vacutainer canister to mitigate needle stick injuries.

These and other features of the concepts provided herein will become more apparent to those of skill in the art in view of the accompanying drawings and following description, which describe particular embodiments of such concepts in greater detail.

DRAWINGS

A more particular description of the present disclosure will be rendered by reference to specific embodiments thereof that are illustrated in the appended drawings. It is appreciated that these drawings depict only typical embodiments of the invention and are therefore not to be considered limiting of its scope. Example embodiments of the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

FIG. 1A illustrates a side view of an intraosseous access system including an aspiration system, in accordance with some embodiments.

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FIG. 1B illustrates a perspective view of an intraosseous access system including an aspiration system, in accordance with some embodiments.

FIG. 2 illustrates a plan, cross-section view of an intraosseous access system including an aspiration system having a syringe, in accordance with some embodiments.

FIG. 3 illustrates a plan, cross-section view of an intraosseous access system including an aspiration system having a vacutainer, in accordance with some embodiments.

FIG. 4 illustrates a flow chart of an exemplary method of use for an intraosseous access system, in accordance with some embodiments.

DESCRIPTION

Before some particular embodiments are disclosed in greater detail, it should be understood that the particular embodiments disclosed herein do not limit the scope of the concepts provided herein. It should also be understood that a particular embodiment disclosed herein can have features that can be readily separated from the particular embodiment and optionally combined with or substituted for features of any of a number of other embodiments disclosed herein.

Regarding terms used herein, it should also be understood the terms are for the purpose of describing some particular embodiments, and the terms do not limit the scope of the concepts provided herein. Ordinal numbers (e.g., first, second, third, etc.) are generally used to distinguish or identify different features or steps in a group of features or steps, and do not supply a serial or numerical limitation. For example, “first,” “second,” and “third” features or steps need not necessarily appear in that order, and the particular embodiments including such features or steps need not necessarily be limited to the three features or steps. Labels such as “left,” “right,” “top,” “bottom,” “front,” “back,” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. Singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

With respect to “proximal,” a “proximal portion” or a “proximal-end portion” of, for example, a needle disclosed herein includes a portion of the needle intended to be near a clinician when the needle is used on a patient. Likewise, a “proximal length” of, for example, the needle includes a length of the needle intended to be near the clinician when the needle is used on the patient. A “proximal end” of, for example, the needle includes an end of the needle intended to be near the clinician when the needle is used on the patient. The proximal portion, the proximal-end portion, or the proximal length of the needle can include the proximal end of the needle; however, the proximal portion, the proximal-end portion, or the proximal length of the needle need not include the proximal end of the needle. That is, unless context suggests otherwise, the proximal portion, the proximal-end portion, or the proximal length of the needle is not a terminal portion or terminal length of the needle.

With respect to “distal,” a “distal portion” or a “distal-end portion” of, for example, a needle disclosed herein includes a portion of the needle intended to be near or in a patient when the needle is used on the patient. Likewise, a “distal length” of, for example, the needle includes a length of the needle intended to be near or in the patient when the needle is used on the patient. The

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distal portion, the distal-end portion, or the distal length of the needle can include the distal end of the needle; however, the distal portion, the distal-end portion, or the distal length of the needle need not include the distal end of the needle. That is, unless context suggests otherwise, the distal portion, the distal-end portion, or the distal length of the needle is not a terminal portion or terminal length of the needle.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art.

FIG. 1A illustrates a side view of an intraosseous (IO) access system (“system”) 100 including an aspiration system 110, in accordance with some embodiments. FIG. 1B illustrates a perspective view of an intraosseous access system 100 with portions thereof shown in wire frame.

The system 100 can generally include an access system 102 and an aspiration system 110. The access system 102 can include a drive train 106 coupled to a needle assembly 160. The drive train 106 can be configured to rotate the needle assembly 160 and drill a needle 162 through a bone cortex to access a medullary cavity. The aspiration system 110 can be in fluid communication with a needle 162 of the needle assembly 160 and can confirm access to the medullary cavity by aspirating bone marrow through a needle lumen 164. Advantageously, the intraosseous access system 100 can be configured to determine when the medullary cavity has been accessed by aspirating the needle 162 without disengaging the needle assembly 160 from the system 100.

In an embodiment, the intraosseous access system 100 can include a housing 104 including one or more of the access system 102, the aspiration system 110, or portions thereof, disposed therein. The access system 102 can include a needle assembly 160 rotatably coupled to the drive train 106 and extending from a distal end of the housing 104. Optionally, the housing 104 can include a grip, e.g. a pistol grip 108, or similar structures, configured to facilitate grasping the housing 104 and urging the needle 162 distally into a bone.

In an embodiment, the drive train 106 of the access system 102 can be configured to provide rotational movement to one or both of the aspiration system 110 and the needle assembly 160, as described in more detail herein. In some embodiments, the access system 102 can be an automated driver (e.g. a drill that achieves high rotational speeds). In an embodiment, the access system 102 can be an automated driver that includes a drive train 106 having an electrical energy source (e.g. battery) to provide electrical power to a drilling mechanism, e.g. an electric motor, or the like. In an embodiment, the system 100 can include an actuator, e.g. button, switch, or the like, configured to actuate the drive train 106. In an embodiment, the drive train 106 can be actuated by an axial pressure applied to a distal tip of the needle 162.

In an embodiment, the drive train 106 can include a spring powered drilling mechanism, e.g. a coiled spring, flat spring, torsion spring, or similar biasing member, which may store potential mechanical energy and may be released upon actuation of the drive train 106. In an embodiment, the drive train 106 can further include one or more gear mechanisms, biasing members, bearings, bushings, or the like, configured to facilitate rotating the needle assembly 160. In an embodiment, the access system 102 can be a manual driver where a clinician can use a handle, pistol grip 108, or similar structure to urge the needle assembly 160 through the bone cortex.

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In an embodiment, a portion of the housing **104** can include the aspiration system **110** disposed therein. The housing **104** can include one or more viewing windows **120** configured to allow a user to observe a portion of the aspiration system **110** disposed therein. In an embodiment, the viewing window **120** can define an aperture extending through a wall of the housing **104**. In an embodiment, the viewing window **120** can include a transparent or translucent portion of the housing **104**. In an embodiment, the entire housing **104** can be formed of a translucent or transparent material configured to allow a user to observe one or both of the aspiration system **110** and the access system **102**, or portions thereof, disposed therein.

As shown in FIGS. 1A-2, in an embodiment, the aspiration system **110** can include a syringe **130** having a barrel **132**, and a plunger **134** slidably engaged therewith. The barrel **132** can be in fluid communication with a lumen **164** of the needle **162**. Sliding the plunger **134** relative to the barrel **132** can create a vacuum within the barrel **132** and can draw a fluid flow through the needle lumen **164** and into the barrel **132**. In an embodiment, a portion of the barrel **132** can be formed of a translucent or transparent material. In use, a user can observe the fluid flow within the syringe barrel **132** through one of the viewing windows **120**.

In an embodiment, the aspiration system **102** can include a vacutainer, or similar structure configured to maintain a vacuum therein, as described in more detail herein. In use, a user can actuate a release mechanism, e.g. actuator, valve, etc., to place the vacutainer in fluid communication with the needle lumen **164** and draw a fluid flow into the vacutainer. A portion of the vacutainer can be formed of a transparent or translucent material, configured to allow a user to observe a fluid flow, as described herein.

In an embodiment, the aspiration system **110** can be coupled to the access system **102**, or portions thereof. In an embodiment, the aspiration system **110** can be coupled to the needle assembly **160**, or portions thereof. In an embodiment, one or both of the aspiration system **110** and the needle assembly **160** can be rotatable relative to the housing **104**. In an embodiment, the drive train **106** can be configured to rotate the needle assembly **160** and aspiration system **102** assembly about a central longitudinal axis **10**.

In an embodiment, a distal end of the syringe barrel **132** can be coupled with the access system **102**, or the needle assembly **160**, or both. In an embodiment, a distal end of the syringe barrel **132** can be integrally formed with the access system **102**. In an embodiment, a distal end of the syringe barrel **132** can be selectively coupled with the access system **102** using a threaded engagement system, an interference fit, press-fit, snap-fit engagement, luer lock, combinations thereof, or the like. In an embodiment, a distal end of the syringe barrel **132** can be coupled with the access system **102** using adhesive, bonding, welding, or the like.

In an embodiment, the syringe barrel **132** can be coupled with the needle assembly **160** and the drive train **106** and can rotate both the barrel **132** and the needle assembly **160** about a central longitudinal axis **10** relative to the housing **104**. Further, the plunger **134** can remain substantially stationary relative to the housing **104** with regards to any rotational movement about the central longitudinal axis **10**. To note, the plunger **134** may still be slidably engaged with the syringe barrel **132** parallel to a longitudinal axis. Further, the plunger **134** can also be slidable along the longitudinal axis **10** relative to the housing **104**.

In an embodiment, the needle assembly **160** can include an obturator **166** disposed within a needle lumen **164** and configured to prevent tissue and bone fragments from enter-

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ing and occluding the needle lumen **164**. A proximal end of the obturator **166** can be coupled to the plunger **134**. In an embodiment, the obturator **166** can be rotatable relative to the plunger **134**. For example, the obturator **166** can be coupled to the plunger **134** using a bearing **150**. As used herein a bearing **150** can include a ball-bearing, bushing, or similar structure configured to facilitate rotational movement about an axis. As such, as the needle assembly **160**, including the needle **162** and the obturator **166**, is rotated by the drive train **106**, the obturator **166** can rotate relative to the plunger **134**. Further, sliding the plunger **134** longitudinally, relative to the housing **104**, can slide the obturator **166** relative to the needle **162**.

In use, rotating the needle assembly **160** can drill a needle **162** through a bone cortex and access the medullary cavity. In an embodiment, once the user believes the medullary cavity has been accessed, the plunger **134** can be withdrawn proximally to both create a vacuum within the barrel **132** and remove the obturator **166** from the needle lumen **164** to place the barrel **134** in fluid communication with the needle lumen **164**. If access to the medullary cavity has been achieved, bone marrow can be aspirated through the needle lumen **164**. A user can visualize bone marrow in the syringe barrel **132** through the one or more viewing windows **120** to confirm access to the medullary cavity has been achieved. Advantageously, the aspiration system **110** allows for confirmation of access to the medullary cavity by bone marrow aspiration without the need to disassemble the needle assembly **160** from the intraosseous access system **100**. In the event that access to the medullary cavity has not been achieved, i.e. no bone marrow is observed, the user can advance the plunger **134** distally, replacing the obturator **166** within the needle lumen **164** and continue drilling the needle assembly **160** through the bone cortex.

FIG. 2 illustrates a plan, cross-section view of an intraosseous access system **100**. In some embodiments, the one or more viewing windows **120** can be located on a distal portion of the housing body **112** and can be configured to allow a user to observe aspiration of fluids from the needle lumen **164**.

In some embodiments, the plunger **132** can include a plunger shaft **146** extending longitudinally through a portion of the syringe barrel **134**. A proximal end of the plunger shaft **146** can be coupled to a plunger handle **142** extending perpendicular thereto. In an embodiment, the plunger handle **142** can be rotatably coupled to the plunger shaft **146** and can include a bearing **150**, disposed therebetween. As such, in an embodiment, the drive train **106** can rotate one or more of the needle assembly **160**, syringe barrel **132**, and plunger shaft **146** about the central longitudinal axis **10**, and the handle **142** can remain substantially stationary with regards to any rotational movement. In an embodiment, a portion of the plunger handle **142**, e.g. a first portion **142A** and a second portion **142B**, can extend from the plunger shaft **146** perpendicular to a longitudinal axis. In an embodiment, a portion of the plunger handle **142** can extend from the plunger shaft **146** perpendicular to a longitudinal axis to contact a wall of the housing **104**.

In an embodiment, a portion of the plunger handle **142** can extend from the plunger shaft **146** perpendicular to a longitudinal axis and extend through a housing aperture **126** disposed in a wall of the housing **104**. In an embodiment, a portion of the plunger handle **142** can engage the housing aperture **126** and stabilize the plunger **134** relative to the housing **104**, e.g. prevent the handle **142** from rotating relative to the housing **104** about the longitudinal axis **10**. For example, a first portion **142A** can extend through a first

housing aperture **126A** and a second portion **142B** can extend through a second housing aperture **126B**. In an embodiment, the housing aperture **126** can define an elongate shape extending longitudinally. In use a user can grasp a portion of the plunger handle **142**, extending through the housing aperture **126** and can urge the plunger **134** longitudinally relative to the housing **104**.

In an embodiment, a distal end of the plunger shaft **146** can be coupled to a piston **148**. The piston **148** can be formed of a compliant material, e.g. rubber or the like, and can engage an inner surface of the barrel **132** to provide a fluid tight seal therebetween. In use, withdrawing the plunger **134** proximally, along a longitudinal axis can create a vacuum between a proximal surface of the piston **148** and a distal end of the barrel **132**.

In an embodiment, the piston **148** can be rotatably coupled to the plunger shaft **146** by way of a bearing **150**, bushing, or similar structure. In use, the drive train **106** can rotate the needle assembly **160**, the syringe barrel **132** and the piston **148** about the longitudinal axis **10** while the plunger shaft **146** and the housing **104** can remain rotationally stationary relative to the longitudinal axis **10**.

In an embodiment, the piston **148** can be in a fixed relationship relative to the plunger shaft **146**. The piston **148** can then be rotatable relative to the syringe barrel **134** while maintaining a fluid tight seal therebetween. In use, the drive train **106** can rotate the needle assembly **160** and the syringe barrel **132** about the longitudinal axis **10** while the plunger shaft **146**, the piston **148**, and the housing **104** can remain rotationally stationary relative to the longitudinal axis **10**.

In an embodiment, an obturator **166** can be disposed within the needle lumen **164**. A proximal end of the obturator **166** can be coupled to the piston **148** of the plunger **134**. In some embodiments, the obturator **166** can be coupled to the piston **148** by a snap-fit, press-fit, or interference fit engagement, or the like, or by adhesive, welding, bonding, combinations thereof, or the like. In an embodiment, the obturator **166** can be coupled in a fixed relationship relative to the plunger **134**. In an embodiment, the obturator **166** can be rotatably coupled to the plunger **134** by way of a bearing **150**, bushing, or similar structure. As such, the obturator can rotate independently of the piston **148**.

In some embodiments, the obturator **166** can be removed from the needle lumen **164** to provide fluid communication, through the needle lumen **164** and into the syringe barrel **132**. In an embodiment, withdrawing the plunger **134** can simultaneously generate a vacuum within the syringe barrel **132** and remove the obturator **166** at least partially from the needle lumen **164**.

In some embodiments, where no fluid flow is observed, the plunger **134** can be advanced distally and the obturator **166** can be replaced back into the needle lumen **164**. In an embodiment, the vacuum within the syringe barrel **132** can facilitate advancing the obturator **166** distally, replacing the obturator **166** back into the needle lumen **164**. The obturator **166** can then continue to prevent occlusion of the needle lumen **164** as the needle assembly **160** continues to penetrate the bone cortex. Advantageously, the obturator **166** can be withdrawn from the needle **162** and into the barrel **132** of the syringe **130**. As such the barrel **132** can prevent accidental needle stick injuries by the distal end of the obturator **166**.

In an embodiment, the drive train **106** can provide rotational movement to the needle assembly **160** in a variety of ways. For example, with the aspiration system **110** being coupled to the needle assembly **160**, the drive train **106** can rotate both the aspiration system **110** and the needle assembly **160**. In some embodiments, the drive train **106** may

rotate only the needle assembly **160**, i.e. without rotating the aspiration system **110**. In some embodiments, the drive train **106** may rotate one or more components of the aspiration system **110** and/or needle assembly **160** while maintaining other components stationary relative thereto.

In an embodiment, the drive train **106** can rotate the barrel **132** of the syringe **130** while maintaining the plunger **134** stationary relative to any rotation about the longitudinal axis **10**. In an embodiment, the plunger handle **142** can engage the housing aperture **126** to prevent the plunger **134** from rotating about the longitudinal axis **10** as the barrel **132** rotates. In an embodiment, the plunger shaft **146** may be coupled to a bearing **150** bushing, or the like, disposed within the piston **148**. The bearing **150** is configured to allow rotational movement of piston **148** around the central axis **10** while allowing the plunger shaft **146** to remain substantially stationary. In some embodiments, the syringe **130** is configured to receive a volume of fluid therein. In some embodiments, the capacity of the syringe maybe between 1 ml and 10 ml. It will be appreciated that other volumes of capacity are also considered.

FIG. 3 illustrates further details of an embodiment of an intraosseous access system **100** including an aspiration system **110** having a vacutainer **400**. The aspiration system **110** can be coupled to the access system **102**, as described herein. In an embodiment, the aspiration system **110** can include a vacutainer **400** configured to maintain a vacuum therein. The vacutainer **400** can include a canister **402** configured to maintain a vacuum therein, and a valve **404** configured to control fluid communication between the needle lumen **164** of the needle assembly **160** and the canister **402** vacutainer **400**.

In an embodiment, the vacutainer **400** can further include a handle **442**. In an embodiment, a portion of the handle **442** can extend through a housing aperture **126**, as described herein. In use, a user can grasp the handle **442** and slide the vacutainer **400** longitudinally relative to the housing **104**. In an embodiment, the vacutainer **400** can include a bearing **150**, bushing, or the like, configured to allow the canister **402** to rotate relative to the handle **442**. In an embodiment, a proximal end of the obturator **166** can be coupled a distal end of the vacutainer **400**. In an embodiment, the vacutainer can further include a bearing **150** configured to allow the obturator **166** to rotate relative to the vacutainer **400**, or portions thereof, e.g. the canister **402** or valve **404**.

In an embodiment, the drive train **106** can rotate the vacutainer **400** and the needle assembly **160** about the central axis **10**. When the user believes the medullary cavity has been accessed, a user can grasp the handle **442** and slide the vacutainer **400** proximally to at least partially withdraw the obturator **166** from the needle lumen **164** of the needle. With the obturator **166** withdrawn, a user can actuate the valve **404** and provide fluid communication between the needle lumen **164** and the canister **402** of the vacutainer **400**. The vacuum within the canister **402** can then draw a fluid flow through the needle lumen **164**. In an embodiment, a portion of the canister **402** can be formed of a transparent material and a fluid flow within the canister **402** can be observed through one or more observation windows **120**.

If the medullary cavity has been accessed, the user can observe bone marrow within the canister **402**. If the medullary cavity has not been accessed, e.g. a distal tip of the needle **162** is still disposed within the bone cortex, no bone marrow will be observed. As such, a user can close the valve **404** and slide the handle **442** distally to replace the obturator **166** within the needle lumen **162** and continue drilling the needle assembly **160** through the bone cortex. Advanta-

geously, withdrawing the vacutainer **400** and obturator **166** assembly proximally can maintain a tip of the obturator **166** within the housing **104** and mitigate accidental needle stick injuries. In an embodiment, the handle **442** can be configured to slide the vacutainer **400** along a longitudinal axis and actuate the valve **404**. In an embodiment, a separate actuator can be configured to actuate the valve **404** and transition the valve **404** between an open and a closed position.

In some embodiments, portions of the aspiration system **110** may be configured to be selectively detachable from the access system **102**. For example, in an embodiment, in the aspiration system **110**, the syringe **130**, the vacutainer **400**, or combinations thereof may be configured to be detachable and interchangeable. In an embodiment, sliding the syringe **130** or the vacutainer **400** can be achieved by manually operating and handle, as described herein. In an embodiment, the system **100** can further include a second drive train configured to be actuated by a user and slide the aspiration system **110**, or portions thereof, along a longitudinal axis.

FIG. 4 illustrates an exemplary method of use **500** for intraosseous access system **100**, including an aspiration system **110**. In an embodiment, the method **500** includes assembling the intraosseous access system **100** (block **502**). In some embodiments, assembling the intraosseous access system **100** includes coupling a needle assembly **106** with an access system **102** of the system **100**. The needle assembly **160** rotatably coupled to the system **100**. In some embodiments, assembling the intraosseous access system **100** includes coupling an aspiration system **110** having a syringe **130** or a vacutainer **400** with the system **100**.

In an embodiment, the method **500** includes actuating the access system **102** to rotate the needle assembly **160**, the aspiration system **110**, or components thereof, to drill a needle **162** through a bone cortex to access the medullary cavity (block **504**). In an embodiment, when a user believes the medullary cavity has been accessed, the user can confirm access to the medullary cavity by withdrawing the obturator **166** from a needle lumen **164** (block **506**). A fluid flow can be drawn through the needle lumen **164** by providing a vacuum in fluid communication with the needle lumen **164** (block **508**). The vacuum can be provided either by withdrawing a plunger **134** from a syringe barrel **132**, or by actuating a valve to place a vacutainer **400** in fluid communication with the needle lumen **164**, or combinations thereof. (block **508**).

In an embodiment, the user can then determine if the medullary cavity has been accessed by observing a fluid flow (block **510**). If medullary cavity access has been confirmed, the user can detach the system **100** from the needle **162** (block **512**). If medullary cavity access is not confirmed, the user replace the obturator **166** within the needle lumen **164** (block **514**) and continue drilling through the bone cortex (return to block **504**).

While some particular embodiments have been disclosed herein, and while the particular embodiments have been disclosed in some detail, it is not the intention for the particular embodiments to limit the scope of the concepts provided herein. Additional adaptations and/or modifications can appear to those of ordinary skill in the art, and, in broader aspects, these adaptations and/or modifications are

encompassed as well. Accordingly, departures may be made from the particular embodiments disclosed herein without departing from the scope of the concepts provided herein.

What is claimed is:

1. An intraosseous access system, comprising:
 - a housing;
 - a needle assembly including a needle extending from a distal end of the housing along a longitudinal axis, and an obturator disposed within a lumen of the needle;
 - an aspiration system disposed within the housing and including one of a syringe or a vacutainer, a proximal end of the obturator coupled to the aspiration system, the aspiration system configured to engage and slide the obturator relative to the needle, the aspiration system further configured to provide a vacuum to the lumen of the needle while remaining engaged with the obturator; and
 - a drive train disposed within the housing and configured to rotate the needle assembly and one of a syringe barrel or a vacutainer canister about the longitudinal axis.

2. The intraosseous access system according to claim 1, wherein the drive train includes one of a biasing member, an electric motor, or a battery.

3. The intraosseous access system according to claim 1, wherein the syringe includes a plunger slidably engaged with the syringe barrel and the housing along the longitudinal axis, the syringe barrel rotatable about the longitudinal axis relative to the plunger.

4. The intraosseous access system according to claim 3, further including a piston rotatably coupled to one of the plunger, the syringe barrel, or the obturator.

5. The intraosseous access system according to claim 3, further including a handle coupled to the plunger, a portion of the handle extending through a wall of the housing.

6. The intraosseous access system according to claim 1, wherein the vacutainer includes a valve configured to control fluid communication between the vacutainer canister and the lumen of the needle to selectively apply the vacuum to the lumen of the needle.

7. The intraosseous access system according to claim 6, wherein the vacutainer further includes an actuator configured to slide the vacutainer canister along the longitudinal axis or to actuate the valve between an open and closed position.

8. The intraosseous access system according to claim 1, wherein the housing further includes an observation window configured to allow a user to observe one of the syringe barrel or the vacutainer canister disposed therebelow.

9. The intraosseous access system according to claim 1, wherein one of the syringe barrel or the vacutainer canister can include a transparent material.

10. The intraosseous access system according to claim 2, wherein the biasing member is one of a flat spring, a coiled spring, or a torsion spring.

11. The intraosseous access system according to claim 1, wherein a distal tip of the obturator is configured to be withdrawn into one of the syringe barrel or the housing to mitigate accidental needle stick injuries.

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